

Disproof of conjecture 00000001070: the three-fold sum threshold in $\mathbb{F}_p$

Submission for conjecture 00000001070

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Abstract

Conjecture 00000001070 asserts that the smallest cardinality $|A|$ of a set $A \subseteq \mathbb{F}_p$ with three-fold sum $3A = \mathbb{F}_p$ equals $\lceil (p+1)/3 \rceil + 1$. We show that this is **false**. The three-element set $A = \{0, 1, 3\} \subseteq \mathbb{F}_7$ already satisfies $3A = \mathbb{F}_7$, while the conjectured expression evaluates to $\lceil 8/3 \rceil + 1 = 4 \neq 3$. We also explain the arithmetic slip: the value $\lceil (p+2)/3 \rceil$, obtained from Cauchy–Davenport, is the **universal** threshold (every k -element set covers $\mathbb{F}_p$ when $k \geq \lceil (p+2)/3 \rceil$), not the **existential** minimum stated in the conjecture; the latter is far smaller and is known only through additive basis theory. The conjecture is wrong under either reading.

1 The conjecture

Definition 1 (Three-fold sumset). Let p be prime and let $A \subseteq \mathbb{F}_p$ be nonempty. The *three-fold sumset* is

$$3A = A + A + A = \{a + b + c : a, b, c \in A\},$$

where repetitions of summands are allowed, and addition is in $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

Conjecture 2 (as filed). The smallest $|A|$ with $3A = \mathbb{F}_p$ is

$$\left\lceil \frac{p+1}{3} \right\rceil + 1.$$

2 The counterexample at $p = 7$

Theorem 3. *Conjecture 2 is false. Explicitly,*

$$A = \{0, 1, 3\} \subseteq \mathbb{F}_7, \quad |A| = 3, \quad 3A = \mathbb{F}_7,$$

while

$$\left\lceil \frac{7+1}{3} \right\rceil + 1 = \lceil 8/3 \rceil + 1 = 3 + 1 = 4 \neq 3.$$

Thus the conjectured expression overestimates the smallest $|A|$ already at $p = 7$.

Proof. The set A has three distinct elements. Enumerating the $3 \times 3 \times 3 = 27$ ordered triples, the distinct three-fold sums are

y	representation as $a + b + c$
0	$0 + 0 + 0$
1	$0 + 0 + 1$
2	$0 + 1 + 1$
3	$0 + 0 + 3$
4	$0 + 1 + 3$
5	$1 + 1 + 3$
6	$0 + 3 + 3$

so $3A = \{0, 1, 2, 3, 4, 5, 6\} = \mathbb{F}_7$. The arithmetic value of the conjectured formula is 4, and $3 < 4$. $\square$

Remark 4. No two-element subset of $\mathbb{F}_7$ has three-fold sum $\mathbb{F}_7$; the $21 = \binom{7}{2}$ two-element subsets each produce at most 4 distinct sums. Hence the true minimum at $p = 7$ is exactly 3, and the conjectured value 4 is not merely unattained but strictly larger than the truth.

3 Why the formula is wrong: two different thresholds

The conjectured expression is motivated by Cauchy–Davenport, but it answers a different question from the one that is stated. We separate the two.

3.1 The universal threshold

Theorem 5 (Cauchy–Davenport). *For nonempty $A, B \subseteq \mathbb{F}_p$ with p prime,*

$$|A + B| \geq \min(p, |A| + |B| - 1).$$

Iterating Theorem 5 twice gives $|3A| \geq \min(p, 3|A| - 2)$:

$$\begin{aligned} |2A| &\geq \min(p, 2|A| - 1), \\ |3A| = |A + 2A| &\geq \min(p, |A| + \min(p, 2|A| - 1) - 1) \geq \min(p, 3|A| - 2). \end{aligned}$$

Consequently, if $3|A| - 2 \geq p$ then $|3A| \geq p$, so $3A = \mathbb{F}_p$ for *every* set A of that size. Writing $k = |A|$, the least k with $3k - 2 \geq p$ is

$$k_{\text{univ}}(p) = \left\lceil \frac{p+2}{3} \right\rceil.$$

This is sharp for the universal statement: for $k < \lceil (p+2)/3 \rceil$ the interval $A = \{0, 1, \dots, k-1\}$ has $3A = \{0, 1, \dots, 3k-3\}$ of size $3k-2 < p$, hence does not cover $\mathbb{F}_p$.

Proposition 6. *The conjectured value $\lceil (p+1)/3 \rceil + 1$ equals the universal threshold $\lceil (p+2)/3 \rceil$ if and only if $p \equiv 2 \pmod{3}$. It is too large by exactly 1 for $p = 3$ and for every prime $p \equiv 1 \pmod{3}$; for $p \leq 30$ these are*

$$p = 3, 7, 13, 19.$$

Proof. Write the remainder of p modulo 3. If $p = 3m+2$, then $\lceil (p+1)/3 \rceil + 1 = m+2 = \lceil (p+2)/3 \rceil$. If $p = 3m+1$, then $\lceil (p+1)/3 \rceil + 1 = m+2$ while $\lceil (p+2)/3 \rceil = m+1$, a difference of 1. If $p = 3$ then $\lceil 4/3 \rceil + 1 = 3$ while $\lceil 5/3 \rceil = 2$, again a difference of 1. $\square$

So even if the intended quantity were the universal threshold, the conjecture would be wrong at $p = 3$ and at every prime $p \equiv 1 \pmod{3}$.

3.2 The existential minimum stated in the conjecture

Conjecture 2 literally asks for the *smallest* $|A|$ for which *some* A covers $\mathbb{F}_p$. This existential minimum is much smaller than the universal threshold, because a k -element set can produce

$$\left| \{a + b + c : a, b, c \in A\} \right| \leq \binom{k+2}{3} = \frac{k(k+1)(k+2)}{6}$$

distinct sums (choosing an unordered multiset of three summands). Covering all p residues therefore forces $\binom{k+2}{3} \geq p$, so $k \gtrsim (6p)^{1/3}$; the true threshold grows like $p^{1/3}$, not linearly in p . Exhaustive search for all primes $p \leq 30$ gives the following values, where “true” is the smallest $|A|$ with some $3A = \mathbb{F}_p$.

p	true (existential)	$\lceil (p+1)/3 \rceil + 1$ (conjecture)	$\lceil (p+2)/3 \rceil$ (universal)
2	2	2	2
3	2	3	2
5	3	3	3
7	3	4	3
11	4	5	5
13	4	6	5
17	5	7	7
19	5	8	7
23	5	9	9
29	6	11	11

Remark 7. For $p \leq 30$ the conjectured value matches the true existential minimum only at $p = 2$ and $p = 5$; at $p = 3, 7, 11, 13, 17, 19, 23, 29$ it is strictly too large. The universal threshold likewise fails to be the existential minimum from $p = 11$ onward (for instance $A = \{0, 1, 2, 4\}$ of size 4 covers $\mathbb{F}_{11}$, while $\lceil 13/3 \rceil = 5$). Thus the conjectured formula is not the answer to either question.

4 Formal verification

The counterexample of Theorem 3 is machine-checked in the accompanying Lean 4 project (`lean4/`). The field $\mathbb{F}_7$ is modelled as `Fin 7` with its cyclic addition; the three-fold sum is a decidable predicate

$$\text{IsThreefoldSum } A \ y : \iff \exists a \ b \ c : \text{Fin } 7, \ a \in A \wedge b \in A \wedge c \in A \wedge a + b + c = y,$$

and the project proves

$$\forall y : \text{Fin } 7, \text{ IsThreefoldSum } \{0, 1, 3\} \ y, \quad |\{0, 1, 3\}| = 3, \quad \lceil 8/3 \rceil + 1 = 4 \neq 3.$$

Only core Lean (`import Std, no Mathlib, no ZMod, no sorry`) is used.

5 Conclusion

Conjecture 00000001070 is **false**. The set $A = \{0, 1, 3\}$ shows $3A = \mathbb{F}_7$ with $|A| = 3$, whereas the conjectured formula returns 4. The underlying identity $\lceil (p+2)/3 \rceil = \lceil (p+1)/3 \rceil + 1$ has the wrong parity of rounding: the two sides agree only when $p \equiv 2 \pmod{3}$, and differ by one at $p = 3$ and all primes $p \equiv 1 \pmod{3}$. Moreover, neither expression is the existential minimum that the conjecture states; that minimum grows like $p^{1/3}$.